

Fever

Learning Objectives

- ⇒ To learn the exact meaning of normal temperature and fever
- ⇒ To learn about core temperature and its mechanism
- ⇒ To learn about the recording of the temperature
- ⇒ To learn about the stages of fever
- ⇒ To learn about the types of fever
- ⇒ To learn about the clinical causes of fever
- ⇒ To learn about the pathophysiology of fever
- ⇒ To know about the clinical features of measles, chickenpox, herpes zoster, hand foot mouth disease, rubella, malaria, filariasis, flu, dengue, leptospirosis, chikungunya, typhoid, other regional disorders with fever
- ⇒ To know about the organisms, incubation period and mode of transmission
- ⇒ To know about common regional disorders with fever complications

The term fever refers to the pathological mechanism where in raised body temperature is seen.

Normal temperature control: The ability of human beings to maintain a stable body temperature depends on the control of heat regulating centre situated in the hypothalamus. Neurons in both the preoptic anterior hypothalamus and the posterior hypothalamus receive two kinds of signals: one from peripheral nerves that transmit the information from warmth/ cold receptors in the skin and the other from the temperature of the blood bathing region. These two types of signals are integrated by the thermoregulatory centre of the hypothalamus to maintain normal temperature. The core temperature is regulated at a set point which differs in individuals, but which is usually between 37.0°C and 37.5°C regardless of race and geography. Body temperature however does not remain absolutely constant throughout the day and all individuals have circadian rhythm with evening temperatures $0.5^{\circ} - 1.0^{\circ}\text{C}$ higher than the recording observed in the morning. This daily rhythm is consistent with each individual and is independent of the environment.

In a neutral temperature environment, the metabolic rate of humans produces more heat than is necessary to maintain the core body temperature at 37.0°C .

Clinical thermometer

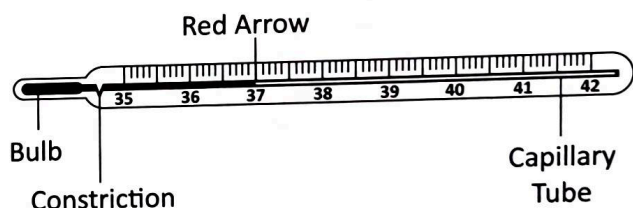


Figure 42: Clinical thermometer with labelling



Figure 43: Infrared clinical thermometer:

- ⊗ A clinical thermometer with Fahrenheit or Celsius scale is used.
- ⊗ The thermometer should be of reliable make and certified as accurate.

- ⊗ It should be kept in situ for a period of 2-3 minutes
- ⊗ The temperature can be recorded oral (keeping the thermometer in the mouth); axillary (in the anterior fold of axillary region); inguinal (in the groin region in case of infants) and rectal (in the rectum). Oral and axillary temperatures are practiced in adults and groin and rectal temperatures are recorded in infants.

Measurement of body temperature:



Figure 44: Temperature gun

- ⊗ While taking the oral reading, the mouth must be closed while breathing should be done through nose. While recording axillary reading, the sweat from the axillary region present in the anterior axillary fold should be wiped out before keeping the mercury bulb of clinical thermometer.
- ⊗ The mercury should be well shaken down before using the thermometer and mercury level should be below the arrow mark of the clinical thermometer.
- ⊗ To determine the type of fever, temperature is recorded every 4-6 hours.

Normal temperature: measurements of oral temperature are about 0.4°C and axillary temperatures upto 1.0°C below core temperature. Both of the recordings are more accurate methods than rectal readings which are affected by heat generated by faecal bacteria.

Pathological mechanism of fever

Types of fever: In the past, much emphasis is being given to determine the pattern of the fever.

- ⊗ If the fever does not touch normal at all, diurnal fluctuations of temperature are within 1.5°F , then such type of fever is termed as continuous fever.
- ⊗ If the fever does not touch normal at all, diurnal fluctuations of temperature exceed 1.5°F , then such

type of fever is termed as remittent fever.

- ⊗ A fever which touches normal for a few hours during the day is called as intermittent, hectic or septic fever
 - Intermittent fever which occurs daily is termed as quotidian fever
 - Intermittent fever which occurs every alternate days is termed as tertian fever
 - Intermittent fever which occurs every 4th day is termed as quartan fever
 - Intermittent fever which does not follow any of the above patterns is termed as irregular fever
 - A fever occurring in bouts lasting for days with afebrile phases is called as relapsing or periodic fever

Stages of fever: in the course of fever, four stages of fever can be recognised.

1. **Prodromal stage:** this stage may be absent
2. **Stage of onset or invasion:** (initial or pyrogenetic phase): associated with initial phase of fever. The onset of fever may be insidious (gradual) or at times step ladder type in case of typhoid or abrupt or sudden as in lobar pneumonia
3. **Stage of full development (stage of fastigium):** full blown symptoms of fever are seen
4. **Stage of decline (stage of defervescence or termination):** subsidence of the symptoms of fever are observed. The termination or fall of fever may be sudden (by crisis) as in pneumonia or gradual (by lysis) as in typhoid fever.

Accompaniments of fever:

- ⊗ A sudden raise in temperature is accompanied by rigor or chills as in adults and by convulsions as in children. Chills or rigors are commonly observed in pyelonephritis, lobar pneumonia, malaria, infective endocarditis.
- ⊗ Recurrent rigors can occur in septicemia, pyemia and bile duct or liver sepsis.
- ⊗ Depending on the area or site of pathological process occurring in the body, the symptoms may vary related to affected systemic function. For example, cough is the accompanying symptom in the person who is having fever due to respiratory involvement.

Clinical causes:

- ⊗ Infections: bacterial, viral, rickettsial, spirochaetal, protozoal, fungal.
- ⊗ Neoplasms
- ⊗ Allergic disorders
- ⊗ Collagen vascular diseases
- ⊗ Collection of pus

- Drugs
- Sarcoidosis or other granulomatous disorders
- Heat fever
- Miscellaneous – inflammatory bowel disease or pulmonary emboli

Mechanism of temperature and fever

Temperature¹ will be raised in the individuals because of two major reasons, namely increased heat production from the body and decreased heat loss from the body. Increased heat generation can occur because of the following reasons.

1. Increased basal rate of metabolism of all cells of the human body
2. Extra rate of metabolism caused by muscular activity, including muscular contractions caused by shivering
3. Extra metabolism caused by the effect of thyroxine
4. Extra metabolism caused by the effect of epinephrine, norepinephrine and sympathetic stimulation on cells
5. Extra metabolism caused by the increased chemical activity inside the cells, especially when the cellular temperature increases
6. Extra metabolism needed for digestion, absorption and storage of food (thermogenic effect of food)

Reduced heat loss from the body: Heat loss from the body occurs in the skin, expired air and urine. As far as physics of heat loss is concerned, loss of heat can occur in three methods radiation, conduction and evaporation.

Radiation: loss of heat by radiation means loss in the form of infrared heat rays, a type of electromagnetic wave. The human body radiates heat rays in all directions. Heat rays are also being radiated from walls of rooms and other objects toward the body. If the temperature of the body is greater than the temperature of the surroundings, a greater quantity of heat is radiated from the body than is radiated to the body.

Conduction: only minute quantities of heat, about 3%, are normally lost from the body by direct conduction from the surface of the body to the solid objects connected to the body such as chair or bed. Loss of heat by conduction of heat to air represents about 15% of the heat loss under normal conditions.

Another way of conduction is convection. This is removal of heat from the body by convection air currents is termed as heat loss by convection. In this way of heat loss, first there will be conduction of heat to the air followed by heat loss by convection air currents.

Evaporation: When water evaporates from the body surface, 0.58 Calorie (kilocalorie) of heat is lost for each

gram of water that evaporates. As per physiologists, water evaporates 600 to 700 ml/ 24 hours from the skin and lungs. This causes continual heat loss at a rate of 16 to 19 Calories/ hour. Anything that prevents adequate evaporation when the surrounding temperature is higher than the skin temperature will cause the internal body temperature to rise.

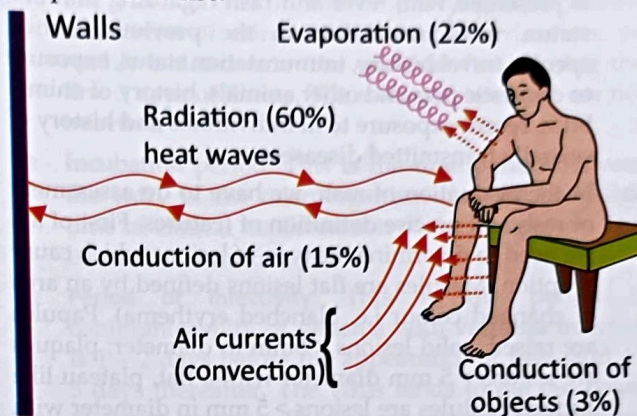


Figure 45: Mechanism of temperature loss from the body

Source: Guyton's physiology, the present image shows the person in disrobed condition, and percentage of heat loss in closed room in radiation, conduction and evaporation.

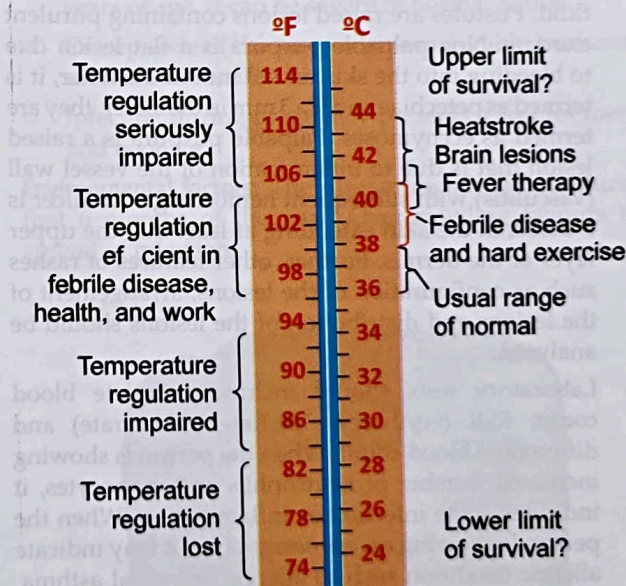


Figure 46: Temperature variations observed in the body

Mechanism of temperature variations under various conditions

Approach to the patient of fever

- Attention must be paid to the chronology of the events and associated symptoms and signs of fever to accurate diagnosis. For example, if the person is

Whenever, there is increase of temperature either due to increased heat production or reduced heat loss, we can infer the vitiation or abnormal increase of Pitta Dosha in the body as primary Dosha affliction. This may occur as a result of increase of Ushna, Tikshna qualities in the body.

presented with headache, running nose, cough and fever, it may indicate the respiratory infection. If the person is presented with epigastric pain, vomiting and fever, it may indicate the upper alimentary canal infection.

- ⊗ A thorough history must be taken when the person is presented with fever and rash regarding immune status, medicines taken in the previous month, specific travel history, immunisation status, exposure to domestic pets and other animals, history of animal bites, recent exposure to ill individuals and history of sexually transmitted diseases.
- ⊗ In the evaluation of rash, we have to do assessment of rash and precise definition of features. First of all, we need to determine the type of lesions which cause eruption. Macules are flat lesions defined by an area of changed colour i.e. blanching erythema). Papules are raised, solid lesions < 5mm in diameter; plaques are lesions > 5 mm diameter with a flat, plateau like surface; nodules are lesions > 5 mm in diameter with a more rounded configuration. Wheals are papules or plaques that pale pink and may appear annular (ring like) as they enlarge and may last only 24–48 hours in any defined area. Vesicles (< 5 mm) and bullae (>5mm) are circumscribed, elevated lesions containing fluid. Pustules are raised lesions containing purulent exudate. Non palpable purpura is a flat lesion due to bleeding into the skin; if < 3 mm in diameter, it is termed as petechiae and if >3 mm in diameter, they are termed as ecchymoses. Palpable purpura is a raised lesion that is due to inflammation of the vessel wall (vasculitis) with subsequent hemorrhage. An ulcer is a defect in the skin extending at least into the upper layer of the dermis. Further, other features of rashes such as configuration of the lesions, arrangement of the lesions and distribution of the lesions should be analysed.
- ⊗ Laboratory tests should include complete blood count, ESR (erythrocyte sedimentation rate) and differential blood count. When the person is showing increased number of neutrophils and monocytes, it indicates acute infection or inflammation. When the person is showing excess eosinophils, it may indicate allergic condition such as allergic bronchial asthma. When the person is showing rapid ESR, it may indicate acute infection. When the person is having ESR more than 100 mm in 1st hour, it may indicate chronic infection such as tuberculosis.
- ⊗ When the person is presented with burning micturition, we have to advise the urinalysis. When the person is presented with cough and yellowish sputum, one has to advise sputum test etc.
- ⊗ When the person is presented with fever persisting for 3 weeks with 1 week of intelligent and invasive investigation, the physician has to repeat the history and physical examination. This status is termed as

FUO (fever of unknown origin) or PUO (pyrexia of unknown origin).

- ⊗ Laboratory testing include in PUO are complete blood count, differential blood count, peripheral smear, ESR, urinalysis, liver function test (LFT), HIV, VDRL, serum creatinine, serum electrolytes. Depending upon the nature of clinical presentation, one has to advise the cultures of blood, urine, sputum and other fluids such as semen, vaginal swab etc. If no proper diagnosis is possible, then one has to go for non invasive techniques such as CT scan and invasive techniques such as needle biopsy or laparoscopy.

Measles

Measles is a highly contagious, viral infection that is most commonly seen in children. This is characterised by fever, cough, running nose, conjunctivitis, enanthem (Koplik's spots) on the buccal or labial mucosa and maculopapular rashes that spread cephalocaudally. Diagnosis is usually clinical.

- ⊗ This is caused by paramyxovirus. This is extremely communicable. This is spread mainly by secretions from the nose, throat and mouth during the prodromal stage or early eruptive stage. Communicability begins several days before and continues until several days after the rash appears. Measles is not communicable once the rash begins to desquamate.
- ⊗ Incubation period: 7 to 14 days; transmitted by respiratory droplets.

Clinical features

- ⊗ Fever, running nose, hacking cough and tarsal conjunctivitis commences during initial phase of the infection.

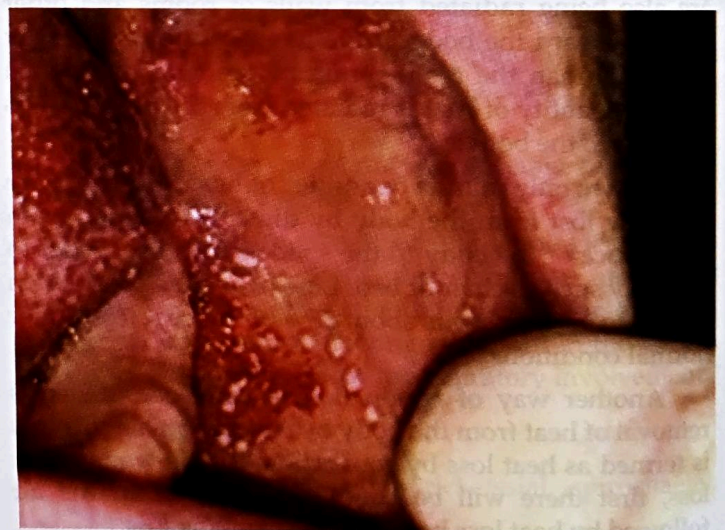


Figure 47: Koplik spots in measles

- ⊗ Koplik's spots (see Figure 47) appear 2 to 4 days later on the buccal mucosa opposite to the 1st and 2nd upper molars. The spots resemble grains of white sand surrounded by red areolae. They may

be extensive, producing diffuse mottled erythema of the buccal mucosa. Sore throat develops during the progression.

- ⊗ The rashes (see Figure 48) appear 3 to 5 days after symptom onset, usually 1 to 2 days after the appearance of Koplik's spots. It begins on the face in front of and below the ears and on the sides of the neck as irregular macules, soon mixed with papules. Within 24 to 48 hours, lesions spread to the trunk and extremities (including palms and soles) as they begin to fade on the face.



Figure 48: Clinical features of measles

- ⊗ During peak disease severity, the patient may have raised body temperature $> 40^{\circ}\text{C}$, with periorbital oedema, conjunctivitis, photophobia, hacking cough, extensive rash and mild itching.
- ⊗ In 3 to 5 days, fever falls, the patient feels more comfortable, and rash fades rapidly with coppery brown discolouration followed by desquamation.
- ⊗ Diagnosis is done by clinical evaluation, serological testing, viral detection via culture or reverse transcription PCR
- ⊗ Differential diagnosis includes rubella, scarlet fever, drug rashes, serum sickness, infectious mononucleosis.

Chicken Pox

Chickenpox or varicella is an acute, highly infectious disease caused by varicella zoster (V-Z) virus. It is characterized by vesicular rashes that may be accompanied by fever and malaise. It is world-wide in distribution and occurs in both epidemic and endemic forms.

Agent factors

- ⊗ Varicella zoster virus also called as human (alpha) herpes virus 3. Primary infections cause chickenpox leading to painful, vesicular, pustular eruption in the distribution of one or more sensory nerve roots.
- ⊗ Source of infection is usually a patient of chickenpox. The virus occurs in the oropharyngeal secretions and lesion of skin and mucosa. The virus can be readily isolated from the vesicular fluid during the first 3 days of illness. The scabs are however are not infective.
- ⊗ Incubation period: This is the time period between the entry of the microorganism into the body and time of clinical manifestation of the symptoms. This is usually 14 to 16 days.
- ⊗ **Period of infectivity:** This is the period of communicability of patients with varicella infection is 1 – 2 days before the appearance of rash, and 4 – 5 days thereafter. The virus tends to die out before the pustular stage. The patient ceases to be infectious once the lesions have crusted.

Host factors

- ⊗ Chickenpox occurs primarily in children under 10 years of age. It can be severe in normal adults.
- ⊗ One episode of chickenpox gives a durable immunity; second attacks are rare.
- ⊗ Infection during pregnancy presents a risk for foetus and neonate.

Environmental factors: The disease mostly occurs during first 6 months of the year. Overcrowding favours the spread of infection.

Clinical features



Figure 49: Clinical features of chicken pox

✱ The clinical spectrum of the chickenpox may vary from a mild illness with only a few scattered lesions to a severe febrile illness with widespread rash. It is divided into two stages: Pre eruptive stage and eruptive stage.

➤ **Pre eruptive stage:** Onset is sudden with mild or moderate fever, pain in the back, shivering and malaise. This stage is very brief, lasting about 24 hours. In adults, the prodromal illness is usually more severe and may last for 2-3 days before the rash comes out.

➤ **Eruptive stage:** In children, the rash is often the first sign. It comes on the day the fever starts. The characters of the rash are

✱ **Distribution** – The rash is symmetrical. It first appears on the trunk where it is abundant, and then comes on the face, arms and legs where it is less abundant. Mucosal surfaces – buccal and pharyngeal are generally involved. Axilla may be affected, but palms and soles are not usually affected. The density of the eruption diminishes centrifugally.

✱ **Rapid evolution** – the rash advances quickly through the stages of macule, papule, vesicle and scab. The first rash which attract attention are often the vesicles filled with clear fluid and looking like dew-drops on the skin. They are superficial in site, with easily ruptured walls and surrounded by an area of inflammation. Usually they are not umbilicated. The vesicles may form crusts without going through pustular stage. Scabbing begins 4-7 days after the appearance of rash.

✱ **Pleomorphism** – A characteristic feature of chickenpox is pleomorphism, i.e. all stages of the rashes (papules, vesicles and crusts) may be seen simultaneously at one time, in the same area. This is due to the rash appearing in successive crops for 4-5 days in the same area.

✱ **Fever** – the fever does not run high but shows exacerbations with each fresh crop of eruption.

Complications of chickenpox

✱ In most cases, chickenpox is a mild, self limiting disease.

✱ In immune compromised patients, severe complications may be manifested. These include hemorrhages, pneumonia, acute cerebellar ataxia, Reye's syndrome (acute encephalopathy with fatty liver).

Diagnosis of chickenpox

Diagnosis of chickenpox is usually clinical. Laboratory diagnosis is seldom required for the confirmation of chickenpox.

Herpes Zoster

Herpes zoster is infection that results when varicella zoster virus reactivates from its latent state in a posterior dorsal root ganglion. Symptoms usually begin with pain along the affected dermatome, followed in 2 to 3 days by a vesicular eruption that is usually diagnostic.

✱ Lancing, dysesthetic or other pain develops in the involved site followed in 2 to 3 days by a rash, usually crops of vesicles on an erythematous base. The site is usually one or more dermatomes in the thoracic or lumbar region. Lesions are typically unilateral (see Figure 50).



Figure 50: Clinical features of herpes zoster

✱ The site where the eruptions developed is usually hyperesthetic and pain may be severe. Lesions usually continue to form for about 3 to 5 days.

✱ Diagnosis is usually clinical.

Hand Foot Mouth Disease

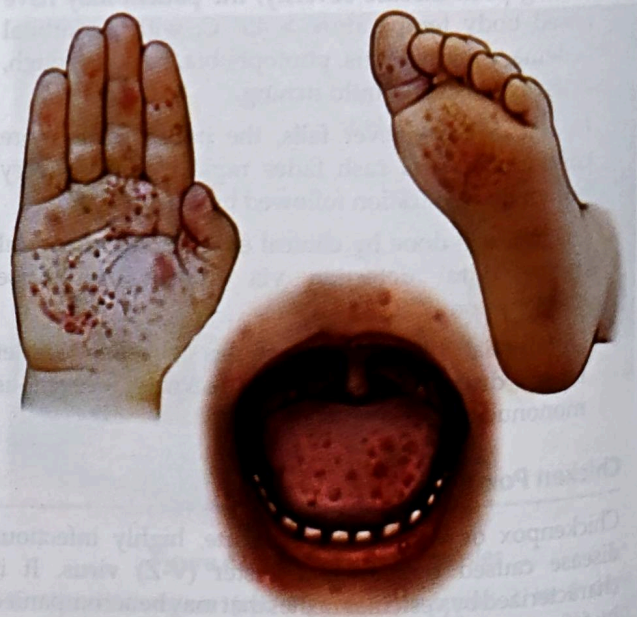


Figure 51: Clinical features of hand, foot and mouth disease

Hand-foot and mouth disease (see Figure 51) is a febrile disorder usually caused by coxsackie virus A16, enterovirus 71 or other enteroviruses. Infection causes a vesicular eruption of skin and mucosa.

- ⊗ The disease is most common in young children.
- ⊗ Children may have sore throat or pain in the mouth and they may refuse to eat. Fever is common.
- ⊗ Vesicles are distributed over the buccal mucosa and tongue, hands and feet and occasionally buttocks and genitals. Usually vesicles are benign and short lived.
- ⊗ Infection with enterovirus 71 may be accompanied by severe neurological manifestations (meningitis, encephalitis, polio like paralysis).
- ⊗ Morbidity and mortality are significantly higher with enterovirus 71 than with coxsackie virus A16 or other enteroviruses.
- ⊗ Diagnosis of the disease is done clinically.

Rubella

- ⊗ Rubella is a contagious viral infection that may cause adenopathy, rash and sometimes constitutional symptoms, which are mild and brief.
- ⊗ Infection during early pregnancy can cause spontaneous abortion, stillbirth and congenital defects. Diagnosis is usually clinical.
- ⊗ Rubella is caused by an RNA virus, rubella virus, which is spread by respiratory droplets through close contact or through air.
- ⊗ Patients can transmit rubella during asymptomatic infection or from 10 days before the appearance of rash until 15 days after onset of rash.
- ⊗ Congenitally infected infants may transmit rubella for many months after birth.
- ⊗ Rubella is less contagious than measles. Immunity appears to be lifelong after natural infection.
- ⊗ Incubation period: 14 to 21 days

Clinical features

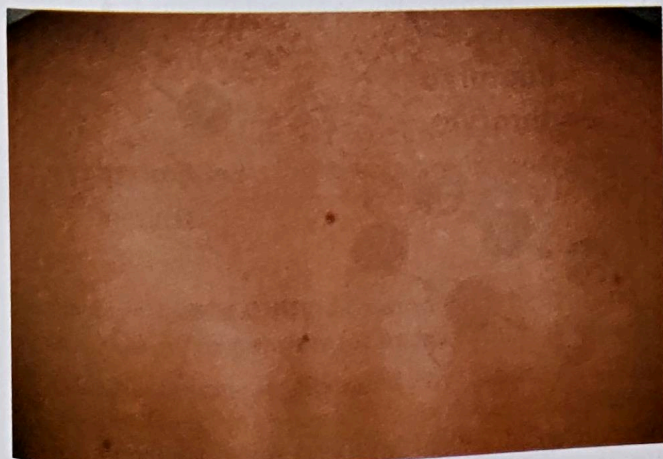


Figure 52: Clinical features of rubella

- ⊗ Initially, the person may have low grade fever, malaise and lymphadenopathy.
- ⊗ Tender swelling of lymph nodes of suboccipital, postauricular and posterior cervical region.
- ⊗ The rash (see Figure 52) is similar to that of measles but is less extensive and more evanescent; it is often the first sign in children.
- ⊗ It begins on the face and neck and quickly spreads to the trunk and extremities.
- ⊗ At onset, a blanching macular erythema may appear, particularly on the face.
- ⊗ On the 2nd day, the rash often becomes more scarlatiniform (pin point) with a reddish flush.
- ⊗ Petechiae form on the soft palate (Forschheimer's spots), later coalescing into a red blush. The rash lasts 3 to 5 days.
- ⊗ Encephalitis may occur in large military outbreaks.
- ⊗ Diagnosis is usually clinical and serological testing may help.
- ⊗ Differential diagnosis includes measles, scarlet fever, secondary syphilis, drug rashes, erythema infectiosum, infectious mononucleosis.

Malaria

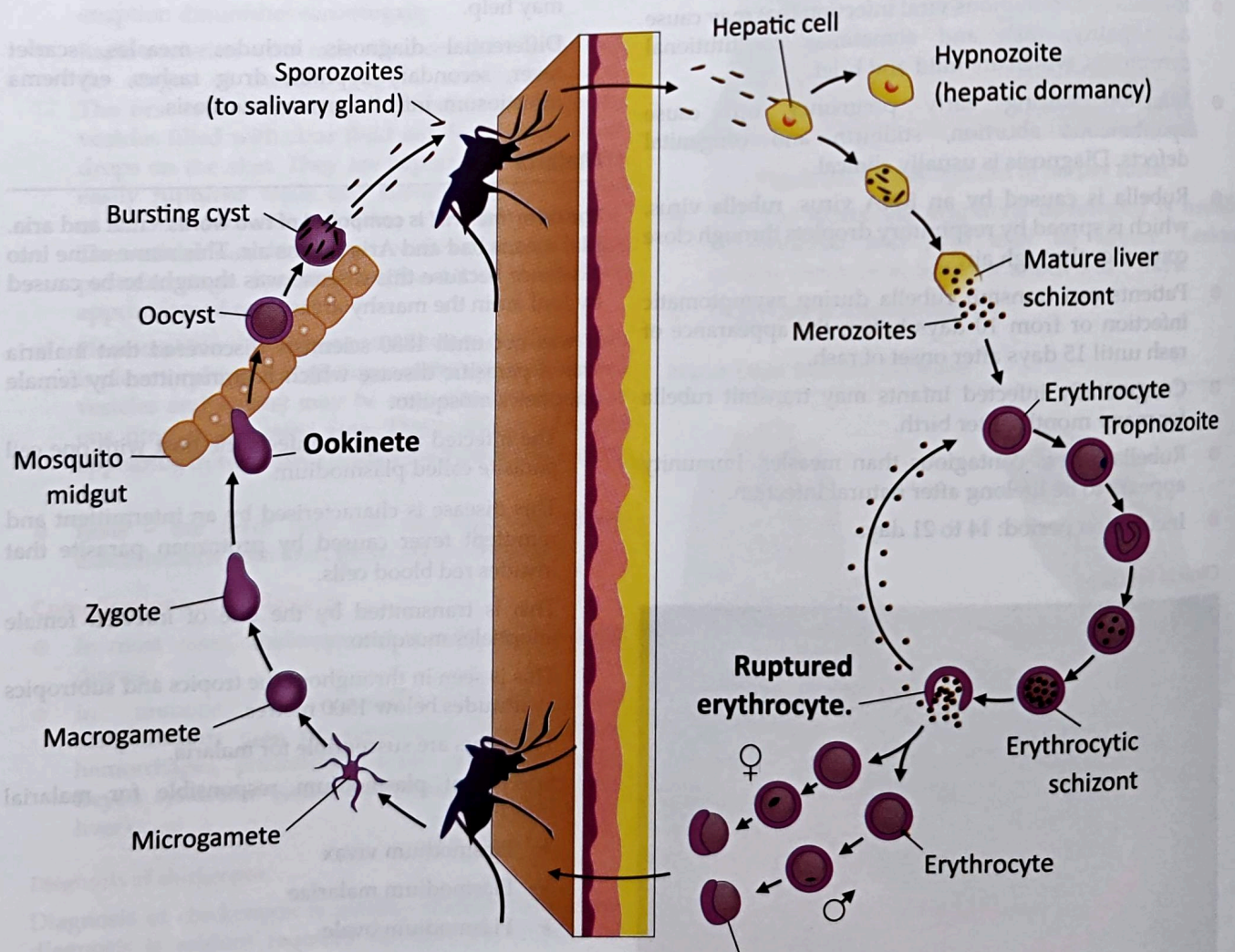
The term 'malaria' is composed of two words – mal and aria. Mal means bad and Aria means air. This name came into existence because this disease was thought to be caused by foul air in the marshy areas.

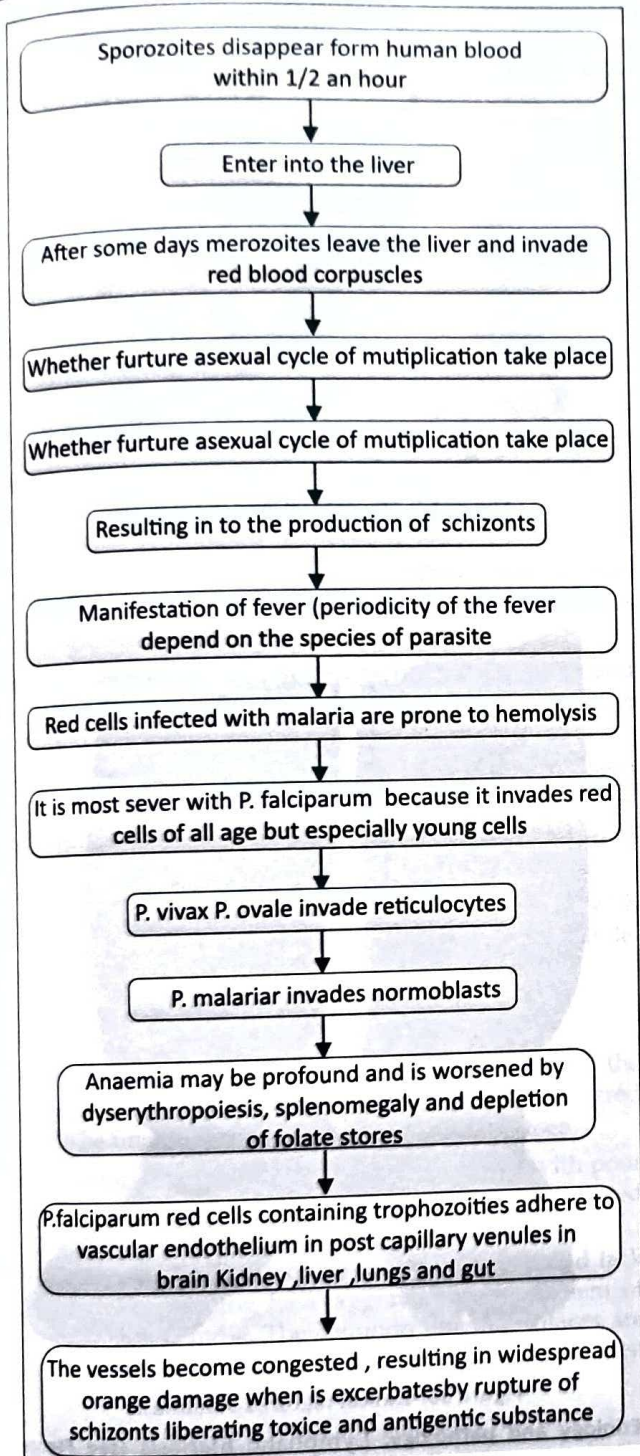
It was not until 1880 scientists discovered that malaria was a parasitic disease which is transmitted by female anopheles mosquito.

- ⊗ The infected mosquito infects the host with one cell parasite called plasmodium.
- ⊗ This disease is characterised by an intermittent and remittent fever caused by protozoan parasite that invades red blood cells.
- ⊗ This is transmitted by the bite of infected female anopheles mosquito.
- ⊗ This is seen in throughout the tropics and subtropics at altitudes below 1500 metres.
- ⊗ Travellers are susceptible for malaria.
- ⊗ Species of plasmodium responsible for malarial infection:
 - Plasmodium vivax
 - Plasmodium malariae
 - Plasmodium ovale
 - Plasmodium falciparum

Distinctive features of individual subtypes of malarial parasites:

Species	Ring form	Tropozoite	Schizont	Gametocyte	Changes in RBCs
<i>P. vivax</i>	Single, occupies 1/3rd of RBC; one chromatin	Irregular in form & size (amoeboid) with vacuole	Larger than RBC; 14-24 merozoites	Round or oval; fills RBC	Enlarged, pale, Schuffner's dots (coarse & pink)
<i>P. malariae</i>	Single, small; one chromatin dot	Band shaped	Smaller than RBC; 5-10 merozoites arranged as a rosette	Similar to those of <i>P. vivax</i> but smaller	Not enlarged; fine pink granules (Zieman's dots)
<i>P. falciparum</i>	More than one ring in a RBC; attached to edges of RBC (eccole or appliquiform) more than one chromatin dot	Not seen in peripheral blood	Not seen in peripheral blood (schizogony occurs in capillaries of internal organs); schizonts with 16-24 merozoites	Crescents (banana shaped)	Normal size; bluish Maurer's dots
<i>P. ovale</i>	Like <i>P. malariae</i> but without distinct bands		Smaller than RBC; round or oval; 8-9 merozoites	Oval, size of RBC and small	Enlarged, oval; Schuffner's dots

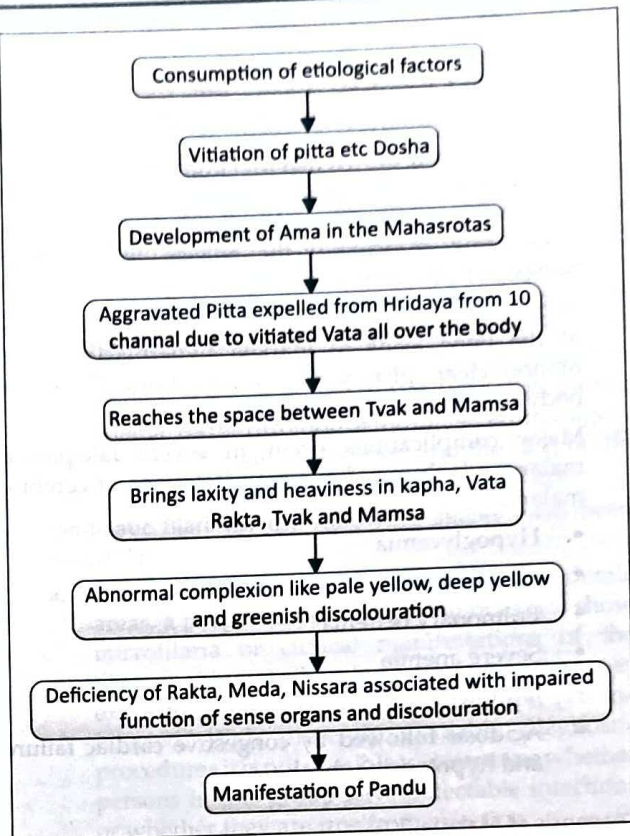
Life cycle of malarial parasite**Figure 53: Life cycle of malaria**



Pathology of Malaria

Morphologic features: Parasitisation and destruction of erythrocytes are responsible for major pathologic changes.

- ❖ Malarial pigment liberated by destroyed RBCs accumulated in phagocytic cells of the reticulo endothelial system resulting in the enlargement of spleen & liver (hepatosplenomegaly).



- ❖ In falciparum malaria, there is massive absorption of haemoglobin by the renal tubules producing black water fever.

- ❖ At autopsy, cerebral malaria is characterised by congestion and petechiae on the white matter.

- ❖ Parasitized erythrocytes in falciparum malaria are sticky and get attached to endothelial cells resulting in obstruction of capillaries of deep organs such as of the brain leading to hypoxia and death.

Initial symptoms:

- Fever
- Headache
- Chills
- Sweats
- Vomiting

Causes of malaria:

- By an infected mosquito
- Blood transfusion
- During pregnancy

Clinical features of Malaria

- ❖ **Cold stage** – Characterised by rigor and headache. The patient feels cold and shivers even though the temperature is raising.

- ⊗ **Fever stage** – the temperature raises to the maximum. The parasites destroy large number of RBCs and thus leading to haemolytic anemia with fever, headache, joint pain, vomiting and diarrhoea.
- ⊗ A characteristic brown malarial pigment, a derivative of haemoglobin is identical to hematin is released from the ruptured RBCs along with the merozoites, discovering principally the spleen, liver, lymph nodes, bone marrow.
- ⊗ Activation of the phagocytic defence mechanisms of the bone leads to marked hyperplasia of the mononuclear phagocytic system throughout the body reflected in splenomegaly.
- ⊗ Major complications occur in severe falciparum malaria which may have manifestations of cerebral malaria.
 - Hypoglycemia
 - Hemoglobinuria
 - Pulmonary oedema
 - Severe anemia
 - Jaundice
 - Acidosis followed by congestive cardiac failure and hypotensive shock

Diagnosis of Malaria

The diagnosis of malaria is made by demonstration of malarial parasite in thin or thick blood films or sometimes in histologic sections.

Laboratory diagnosis of malaria

- ⊗ Microscopic examination of stained blood smears
- ⊗ Antigen capture test
- ⊗ QBC tube test
- ⊗ Newer diagnostic tests which include gene amplification by PCR
- ⊗ Immunological diagnosis

Filariasis

Introduction: Lymphatic filariasis is the infection with any of 3 species of Filarioidea. These species are *Wuchereria bancrofti*, *Brugia malayi* and *B. timori*. Acute symptoms of the infection include fever, lymphadenitis, funiculitis and epididymitis. Chronic symptoms include abscesses, hyperkeratosis, polyarthritis, hydrocele, lymphedema and elephantiasis (see Figure 54).

Tropical pulmonary eosinophilia with bronchospasm, fever and pulmonary infiltrates is another manifestation of infection. Diagnosis is by detection of microfilariae in the blood, ultrasound visualisation of adult worms or serologic testing.



Figure 54: Clinical features of filariasis



Figure 55: Clinical features of filariasis

Etiology and pathology: Lymphatic filariasis (see Figure 55) is caused by 3 species – *Wuchereria bancrofti*, *Brugia malayi* and *B. timori*. Transmission of the microorganisms is by mosquitoes. Infected larvae from the mosquito migrate to the lymphatics, where they develop into adult worms within 6 to 12 months.

Agent factors

- ⊗ The microfilaria of *Wuchereria bancrofti*, *Brugia*

malayi occurring in India display a nocturnal periodicity i.e. they appear in large numbers at night and retreat from the blood during the day. This is a biological adaptation to the nocturnal biting habits of mosquitoes. The maximum density of microfilaria in blood is reported between 10 pm to 2 am.

- ❖ Man is the definite host and mosquito is the intermediate host. The adult worms are usually found in the lymphatic system of the man. The male worms are 40 mm long and female worms are 50 to 100 mm long. The female worms give birth to 50,000 microfilaria a day which find their way into blood circulation via lymphatics. The life span of microfilaria is not known. The adult worms may survive for 15 years or more.

Host factors

- ❖ Man is considered as natural host. All ages are susceptible to infection. In endemic areas, filarial infection has been found even in infants aged less than 6 months.
- ❖ Infection rates raise with age upto 20-30 years and then level off. The infection rates decline in middle and old age.
- ❖ In endemic areas, the disease is more commonly seen in males.
- ❖ The man may get immunity only after many years of exposure.
- ❖ Lymphatic filariasis is often associated with urbanisation, industrialisation, migration of people, illiteracy, poverty and poor sanitation.

Environmental factors

- ❖ **Climate:** climate is a very important factor in the spreading of infection. The rainy season is considered to be time of season of breeding of mosquitoes.
- ❖ **Drainage:** lymphatic filariasis is associated with poor drainage. The vectors breed profusely in polluted water.
- ❖ **Town planning:** Inadequate sewage disposal and lack of town planning have aggravated the problem of filariasis in India. The common breeding places are soakage pits, ill-maintained drains, septic tanks, open ditches, burrow pits etc.

Mode of transmission

Filariasis is transmitted by the bite of infected vector mosquitoes. The parasite is deposited near the site of puncture. It passes through the puncture skin or may penetrate the skin on its own and finally reach the lymphatic system.

The incubation period is the time period of the entry of the microorganism into the body and time of manifestation of the clinical features. In filariasis, the incubation period

is most commonly between 8 to 16 months.

Symptoms and signs

Depend upon the time of entry of the causative organism into the body and the response of the body to the microorganism and duration of the entry of the microorganism into the body. The clinical manifestations can be divided into two distinct clinical types. Lymphatic filariasis caused by the parasite in the lymphatic system and occult filariasis caused by an immune hyperresponsiveness of the human host for e.g. tropical pulmonary eosinophilia. As the microorganism resides in the lymphatics, the person will suffer from non pitting type of oedema. As the duration of the illness is prolonged, the skin becomes thicker and thicker and the skin of the patient becomes like that of the skin of the elephant.

1. Lymphatic filariasis: the following stages have been described.

- Asymptomatic amicrofilaraemia:** In all endemic areas, a proportion of population does not show microfilaria or clinical manifestations of the disease although they have the same degree of exposure to infective larvae as those who become infected. With presently available diagnostic procedures it is not possible to determine whether persons in this group have detectable infections or whether they are free from infection.
- Asymptomatic microfilaraemia:** A considerable proportion of people are asymptomatic, although their blood is positive for microfilaria. They may remain without any symptoms for source of infection in the community. These carriers are usually detected by night blood examination.
- Stage of acute manifestation:** In the first months and years there are recurrent episodes of acute inflammation in lymph glands and vessels. The clinical manifestations comprise filarial fever, lymphangitis, lymphadenitis, lymphoedema of the various parts of the body and epididymoorchitis in the male.
- Stage of chronic obstructive lesions:** The chronic stage usually develops 10-15 years from the onset of the first acute attack. This phase is due to fibrosis and obstruction of lymphatic vessels causing permanent structural changes.

In chronic Bancroftian filariasis, the main clinical features are hydrocele, elephantiasis and chyluria. Elephantiasis may affect the legs, scrotum, arms, penis, vulva and breasts usually in the order of decreasing frequency.

The Brugian filariasis is generally similar to Bancroftian filariasis, but strangely the genitalia are rarely involved.

- Occult filariasis:** The term occult or cryptic filariasis refers to filarial infections in which the classical clinical manifestations are not present and microfilaria are not found in the blood. Occult filariasis is believed

to result from a hypersensitivity reaction to filarial antigens derived from microfilaria. The best example is tropical pulmonary eosinophilia.

- ⊗ **Tropical pulmonary eosinophilia** – is an uncommon manifestation with recurrent bronchospasm, transitory lung infiltrates, low grade fever and marked eosinophilia. This is mostly due to hypersensitivity reaction to microfilariae.

Complications: Chronicity of the pathological process causes enormous thickening and induration of the skin of the legs and scrotum resembling the hide of the elephant and hence the name elephantiasis. **Chylous ascites, chyluria** (see Figure 56) and chylothorax may occur due to the rupture of lymphatics of the abdomen and thorax.

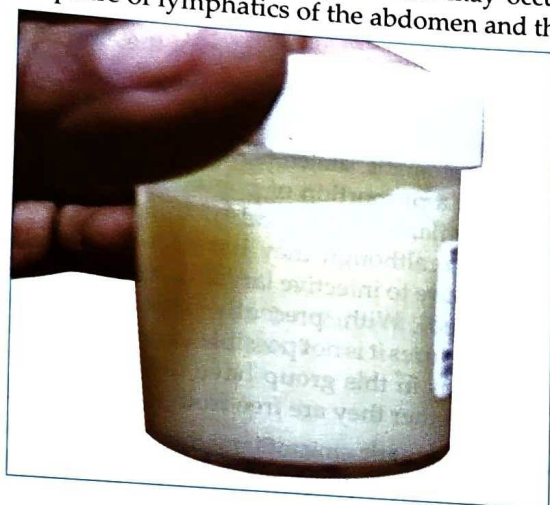


Figure 56: Image of chyluria

Diagnosis

Diagnosis of filariasis is by microscopic examination of blood samples. While doing the peripheral smear examination, detection of microfilariae will confirm the diagnosis of filariasis.

Flu

This is also being termed as influenza. This is a viral, respiratory infection causing fever, running nose, cough, headache and malaise.

Mortality is possible during seasonal epidemics, particularly in high risk patients (who are hospitalised, at the extremes of age, have cardiopulmonary insufficiency or in late pregnancy); during pandemics, even healthy, young patients may die. Diagnosis is usually clinical and depends on local epidemiological patterns.

- ⊗ This is caused by influenza virus. These are classified as types A, B or C by their nucleoproteins and matrix proteins.

Incubation period: ranges from 1 to 4 days with an average of about 48 hours.

- ⊗ In mild cases, many symptoms are like common cold

(sore throat, rhinorrhoea); mild conjunctivitis may occur.

- ⊗ Typical influenza in adults is characterised by sudden onset of chills, fever, prostration, cough and generalised aches and pains (especially in the back and legs). Headache is prominent, often with photophobia and retrobulbar aching.
- ⊗ Respiratory symptoms may be mild at first, with scratchy throat, substernal burning, nonproductive cough, sometimes running nose. Later lower respiratory tract symptoms become prominent.
- ⊗ After 2 to 3 days, the symptoms rapidly subside, although fever may persist for upto 5 days. Cough, weakness, sweating, fatigue may persist for several days or occasionally for weeks.
- ⊗ Complications may appear such as pneumonia which is characterised by worsening cough, blood mixed sputum, dyspnoea, crepitations. Secondary bacterial pneumonia is suggested by persistence or recurrence of fever and cough after primary illness appears to be resolving. Encephalitis, myocarditis and myoglobinuria, sometimes with renal failure develop infrequently after influenza A or B infection.
- ⊗ Diagnosis is by clinical evaluation and sometimes rapid diagnostic testing, pulse oximetry and chest x ray.

Dengue

The word dengue is derived from African word 'denga' meaning fever with haemorrhages. Dengue is mosquito borne disease caused by flavivirus. Dengue fever is usually results in abrupt onset of high fever, headache, myalgia, arthralgia and lymphadenopathy followed by a rash that appears with a 2nd temperature rise after afebrile period. Respiratory symptoms such cough, sore throat and rhinorrhoea can occur.

Dengue can also cause fatal hemorrhagic fever with bleeding tendency and shock. Dengue virus infects blood monocytes, lymphocytes, endothelial cells. This initiates complement activation and consumptive coagulopathy including thrombocytopenia. The entire process takes place rapidly and may evolve over a period of few hours. If the patient is treated appropriately at this stage, there is rapid and dramatic recovery. But, in untreated cases, dengue shock syndrome develops and death occurs.

Epidemiology

The disease spread by the bite of *Aedes aegypti* mosquitoes. The virus circulates in the blood of the infected humans for 2 to 7 days.

Incubation period: 3 to 15 days.

Clinical features

- ⊗ Fever, chills, headache, retro-orbital pain with eye

movement, lumbar backache and severe prostration begins abruptly.

- ⊗ Extreme aching in the legs and joints occurs during the first hours accounting for the traditional name of break bone fever. The temperature rises rapidly upto 40°C with a relative bradycardia.
- ⊗ Fever and other symptoms may persist 48 to 96 hours, followed by rapid defervescence with profuse sweating. Patients then feel well for about 24 hours after which fever may occur again (saddle back pattern), typically with a lower peak temperature than the first. Simultaneously, a blanching maculopapular rash spreads from the trunk to the extremities and face.

Diagnosis

- ⊗ Evaluation should rule out alternative diagnoses, malaria and leptospirosis in particular.
- ⊗ Is based on serologic testing and PCR

Dengue Hemorrhagic Fever

Dengue hemorrhagic fever is a variant presentation that occurs primarily in children < 10 years living in areas where dengue is endemic.

Clinical features

In adults, dengue hemorrhagic fever with abrupt fever and headache, and is initially indistinguishable from classic dengue. Shock and increasing illness may develop rapidly 2 to 6 days after onset. Bleeding tendencies occur usually as purpura, petechiae or ecchymoses at injections sites; sometimes as hematemesis, melena or epistaxis and in some individuals, there will be subarachnoid hemorrhage. Hepatomegaly as well as bronchopneumonia with or without bilateral pleural effusions are common.

Diagnosis is clinical and laboratory criteria. CBC, coagulation tests, urinalysis, liver function tests, dengue serological tests are done to confirm the diagnosis and predict prognosis of the disease.

Leptospirosis

Leptospirosis is an infection caused by one of the several pathogenic serotypes of leptospira. Symptoms are biphasic. Both phases involve acute febrile episodes; the 2nd phase sometimes include haptic, renal and meningeal involvement. Diagnosis is done by darkfield microscopy, culture and serologic testing.

This is a zoonosis occurring in many domestic and wild animals, may cause iapparent illness or serious, even fatal disease.

Human infections are acquired by direct contact with infected urine or tissue or indirectly by contact with contaminated water or soil. Abraded skin and exposed

mucous membranes (conjunctival, nasal, oral) are usual entry portals. Dogs and rats are common probable sources of leptospirosis.

Incubation period: 2 to 20 days (usually 7 to 13 days)

Clinical features

- ⊗ The disease is biphasic.
- ⊗ The septicemic phase starts abruptly, with headache, severe muscular aches, chills, fever, cough, chest pain and in some cases, haemoptysis. Conjunctival suffusion usually appears on the 3rd or 4th day. Splenomegaly and hepatomegaly are uncommon. This phase lasts 4 to 9 days with intermittent nature of occurrence of fever.
- ⊗ Immune phase occurs between 6th and 12th day of the illness, correlating with appearance of antibodies in serum. Fever and earlier symptoms recur, and meningitis may develop.
- ⊗ Weil's syndrome (icteric leptospirosis) is a severe form with jaundice and usually azotemia, anaemia, diminished consciousness and continued fever. Haemorrhagic manifestations which are due to capillary injury and include epistaxis, petechiae, purpura and ecchymoses may be developed. Thrombocytopenia may occur. Signs of hepatocellular and renal dysfunction appear from 3rd to 6th day. When leptospirosis appears with jaundice, mortality rate is 5 to 10%, otherwise, mortality is nil in leptospirosis without jaundice.
- ⊗ **Diagnosis:** blood cultures and serologic testing

Chikun Gunya

Chikungunya is a acute febrile illness followed by more chronic polyarthrititis. It is transmitted by Aedes mosquito and is common in Africa, Asia, Southeast Asia and limited parts of Europe. Prevention involves avoiding the bite of mosquito.

Aetiology

- ⊗ It is an alpha virus that causes fever, rashes and arthropathy.
- ⊗ It is found principally in Africa and Asia including India.
- ⊗ Human and non-human primates are the main reservoirs and the vector for the transmission of the infection is Aedes aegypti mosquito.
- ⊗ Cases occur in epidemics on a background of sporadic cases.

Incubation period : The incubation period of the disease is 2 – 12 days.

Clinical features

- ⊗ A period of fever may be followed by an afebrile

phase and then recrudescence of fever.

- Children may develop maculo-papular rashes.
- Adults are susceptible to arthritis, which causes early morning pain and swelling of the joints and most often observed in small joints.
- Arthritis can persist for months and may become chronic in HLA-B27 positive individuals.
- Related alpha virus causing similar syndromes include Sindbis virus (Scandinavia and Africa), O'nyong-nyong virus (Central Africa), Ross River virus (Australia) and Mayaro virus (Caribbean and South America).
- After its mysterious emergence, the virus virtually disappeared, leaving only occasional evidence of its persistence in Kenya until a transient resurgence of epidemic activity in 1997.

Diagnosis

- It is done by serology but cross reactivity between alpha viruses occurs.
- The rash is most intense on the trunk and limbs and may desquamate.
- Petechiae are occasionally seen, and epistaxis is not uncommon, but this virus is not a regular cause of the HF syndrome, even in children.
- Full blood count: A few patients may develop leukopenia and mildly decreased platelet counts
- LFT: elevated levels of aspartate aminotransferase (AST) have been described.
- CRP (C Reactive Protein): Elevated levels of CRP is seen.

TYPHOID

Typhoid fever is a systemic disease caused by *Salmonella typhi*. Symptoms are high fever, prostration, abdominal pain and a rose coloured rash. Diagnosis is clinical and confirmed by culture.

Epidemiology

- Typhoid bacilli are shed in stool or urine of people with active disease. Inadequate hygiene after defecation may spread *S. typhi* to community food or water supplies. Bacilli will be transmitted more frequently by water than food.
- Flies may spread the organisms from feces to food.
- Occasional transmission by direct contact (feco-oral route) may occur in children during play and in adults during sexual practices.
- The organism enters the body via gastrointestinal tract and gains access to blood stream via the lymphatic channels. Intestinal ulceration, hemorrhage and perforation may occur in severe cases.

Incubation period: 8 – 14 days. The incubation period is inversely related to the number of organisms ingested.

Pathology

Clinical features

- Onset is usually gradual, with fever, headache, arthralgia, pharyngitis, constipation, anorexia, abdominal pain and tenderness are observed. Less common symptoms include dysuria, non productive cough and epistaxis.

Image of pathology of typhoid fever:

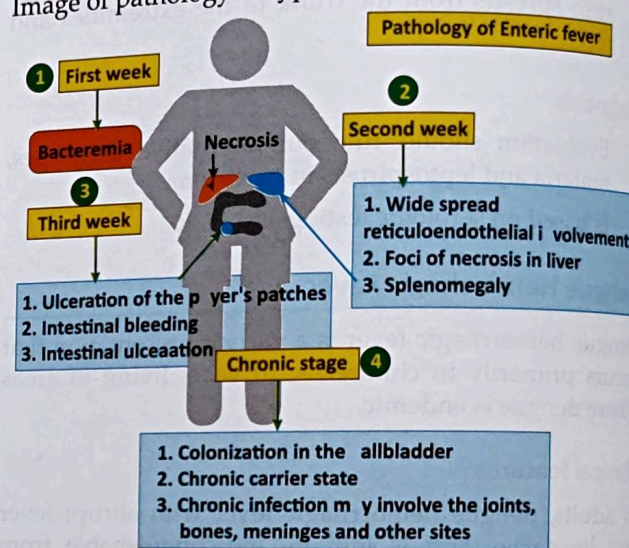


Figure 57: Algorithm of enteric fever

- Without treatment, the temperature raises in steps over 2 to 3 days, remains elevated for another 10 – 14 days. This will reduce gradually at the end of 3rd week and reaches normal levels during 4th week.
- Prolonged fever is often associated with relative bradycardia and prostration.
- In 10% of the patients, discrete pink, blanching lesions (rose spots) appear in crops on the chest and abdomen during 2nd week and resolve in 2 to 5 days.
- Splenomegaly, leukopenia, anemia, liver function dysfunction, proteinuria, acute cholecystitis may occur.
- An acute abdomen and leukocytosis in the 3rd week of presentation, may suggest intestinal perforation.
- Pneumonia may develop during 2nd or 3rd week and may be due to secondary pneumococcal infection.

Diagnosis

- Widal test
- Cultures of blood, stool and urine should be obtained. Blood cultures are usually positive during first 2 weeks of illness, but stool cultures are positive

during 3rd to 5th week. If these cultures are negative and typhoid fever is strongly suspected, culture from a bone marrow biopsy specimen may reveal the organism.

Prognosis

- Without antibiotics, the mortality rate is about 12%, with prompt therapy, the mortality rate is 1%. Most deaths occur in malnourished people, infants and elderly.
- Stupor, coma or shock reflects severity of the disease and poor prognosis of the disease in the affected individuals.

Other Regional Disorders With Fever

Man is surrounded by millions of microorganisms. Some of the microbes are harmless and some are producing more harm to the body. These microorganisms will enter to the body by various routes i.e. mouth, ears, nose, eyes, urethra, anus, vagina, breasts. These microbes may also enter to the body through cuts or injuries occurring to the skin and mucous membranes. When these organisms come in contact with body defence systems, then due to the antigen antibody reaction, the person is likely going to have fever sooner or later depending upon the human response to these organisms.

Four levels of defense will come into play.

1. Tissue macrophages: within minutes after the entry of the microorganism, inflammation begins, the macrophages already present in the tissues, whether histiocytes in the subcutaneous tissues, alveolar macrophages in the lungs, microglia in the brain or others, immediately begin their phagocytic actions. This comes into action within one hour or so.
2. Neutrophil invasion of the inflamed area is a second line of defense. Within the first hour or so after

inflammation, large numbers of neutrophils begin to invade the inflamed area from the blood. This is caused by products from the inflamed tissues that initiate margination (neutrophils to stick to inner surface of capillary endothelium); diapedesis (passage of neutrophils from the capillaries to the tissue of damage); chemotaxis (inflammatory products leading to the movement of neutrophils towards the interstitial spaces). This will lead to neutrophilia (4 to 5 fold increase of number of neutrophils in the body).

3. Second macrophage defense is the third line of defense. Along with the invasion of neutrophils, monocytes from the blood enter the inflamed tissue and enlarge to become macrophages. However, the number of monocytes in the circulating blood is low, also the storage pool of monocytes in the bone marrow is much less than that of neutrophils. Therefore the buildup of macrophages in the inflamed tissue area is much slower than that of neutrophils, requiring several days to become effective. Furthermore, even after invading the inflamed tissue, monocytes are still immature cells, requiring 8 hours or more to swell to much larger sizes and develop tremendous quantities of lysosomes; only then do they acquire the full capacity of tissue macrophages for phagocytosis. After several days to several weeks, the macrophages finally come to dominate the phagocytic cells of inflamed area because of increased bone marrow production of new monocytes.
4. Increased production of granulocytes and monocytes by the bone marrow is the 4th line of defense: This occurs from the stimulation of granulocytic and monocytic progenitor cells of the bone marrow. However, it takes 3 to 4 days before newly formed granulocytes and monocytes to reach the stage of leaving the bone marrow. If the stimulus from the inflamed tissue continues, the bone marrow can continue to produce these cells in tremendous quantities for months and even years, sometimes at a rate 20 to 50 times normal.

Key summary of the chapter

- Any of the metabolic activity happening inside the body will lead to the generation of heat. If metabolic activity increases, this will lead to the increased generation of heat. The person may present with occurrence of fever when there is increased heat production inside the cells or reduced heat loss from the body or both mechanisms occur in an individual.
- Temperature of the body is recorded by clinical thermometer. Nowadays, digital thermometers are used. The thermometer should be kept in oral cavity or axillary region or groin or rectal region. This should be kept in situ for a period of 2 to 3 minutes. To identify the type of fever, one should record the temperature for every 4 to 6 hours.
- Stages of fever: 4 stages are being identified: stage of prodromal stage, stage of onset, stage of full development and stage of decline.
- Depending upon the nature of pathological events taking place in the body, the person may have presenting complaints along with fever, chills or rigor. For example, if the person is having urinary infection, the person may have burning micturition along with fever.
- Inflammation, infection, collection of pus, collagen diseases, allergic diseases, drugs, heat stroke are the probable clinically identifiable causes of fever occurrence.

Self-Assessment: Questions for Reviews

Short essays

- What are the clinical causes of fever occurrence?
- Define fever and explain the stages of fever.
- Describe measles
- Describe chickenpox
- Explain dengue fever
- Explain filariasis
- Describe typhoid
- Describe chikungunya
- Explain leptospirosis
- Explain flu

MCQs

1. What is the amount of time required for the clinical thermometer should be kept to record the body temperature?
 - a. 5 to 6 minutes;
 - b. 1 to 5 minutes;
 - c. 2 to 3 minutes;
 - d. 4 to 5 minutes

Answer: (c)

2. Which of the following regions of the body where clinical thermometer should not be kept?
 - a. Oral cavity;
 - b. Axillary region;
 - c. Rectum;
 - d. Palms

Answer: (d)

3. What is the recording of temperature in Fahrenheit scale when the person is having 37° C on clinical thermometer recording?
 - a. 95° F;
 - b. 98.6° F;
 - c. 101° F;
 - d. 91° F

Answer: (b)

4. What is the formula used for the calculation of temperature to covert from Fahrenheit to Celsius scale?
 - a. $9/5 \times (F-32)$;
 - b. $5/9 \times (F-32)$;
 - c. $9/5 \times (F+32)$;
 - d. $5/9 \times (F+32)$

Answer: (b)

5. Quotidian fever is a type of
 - a. Continuous fever;
 - b. Remittent fever;
 - c. Intermittent fever;
 - d. Irregular fever

Answer: (c)

6. The term quartan fever is used in an individual who is having intermittent fever occurs every
 - a. 6th day;
 - b. Alternate day;
 - c. 4th day;
 - d. Day

Answer: (c)

7. When a clinician makes the diagnosis of PUO (pyrexia of unknown origin) with unidentifiable cause, the duration of presence of fever will be for
 - a. 7 days;
 - b. 10 days;
 - c. 14 days;
 - d. 21 days

Answer: (d)

8. Which of the following is an example for lysis type of subsidence of fever as per western literatures?
 - a. Pneumonia;
 - b. Tuberculosis;
 - c. Typhoid;
 - d. Perforation of gastric ulcer

Answer: (c)

9. Which of the following does not indicate inflammatory events of the body?
 - a. Leucocytosis;
 - b. Leucocytopenia;
 - c. Anemia;
 - d. Polycythemia

Answer: (a)

10. In which part of the oral cavity, clinical thermometer should be kept for recording the temperature of the body?
 - a. Dorsal surface of the tongue;
 - b. Ventral surface of the tongue;
 - c. In between third molar and oral mucosa;
 - d. In between incisors and oral mucosa

Answer: (b)

11. Which of the following is not a viral disease?
 - a. Rubella;
 - b. Measles;
 - c. Chikungunya;
 - d. Typhoid

Answer: (d)

12. Which of the following is a viral disease?
 - a. Malaria;
 - b. Typhoid;
 - c. Filariasis;
 - d. Chickenpox

Answer: (d)

13. Cerebral malaria is caused by
 - a. P. vivax;
 - b. P. falciparum;
 - c. P. ovale;
 - d. P. malariae

Answer: (b)

14. Skin eruptions observed in skin in dermatomal presentation and seen in unilateral presentation is seen in

- a. Chickenpox;
- b. Hand foot mouth disease;
- c. Herpes zoster;
- d. Filariasis

Answer: (c)

15. Non pitting type of oedema is seen in which of the following disease condition?

- a. Chickenpox; b. Filariasis;
- c. Herpes zoster; d. Typhoid

Answer: (b)

16. Filariasis is not caused by

- a. Wuchereria bancrofti;
- b. Brugia malayi;
- c. B. timori;
- d. Vibrio cholerae

Answer: (d)

17. Oedema observed in the patient of filariasis is

- a. Pitting type of oedema;
- b. Non pitting type of oedema;
- c. Both a and b;
- d. Oedema is not observed

Answer: (b)

18. Fever observed in the patient of filariasis is in

- a. Day time;
- b. Night time;
- c. Both a or b;
- d. Both a and b

Answer: (b)

19. The vector responsible for the transmission of filariasis is

- a. Female anopheles mosquito;
- b. Culex mosquito;
- c. Aedis agypti mosquito;
- d. Common fly

Answer: (b)

20. Which of the following is helpful in detection of filariasis?

- a. Blood evaluation;
- b. Urine evaluation;
- c. Sputum evaluation;
- d. Semen evaluation

Answer: (a)

21. Typhoid fever isinfection.

- a. Bacterial;

- b. Viral;
- c. Fungal;
- d. Protozoan

Answer: (a)

22. Incubation period of typhoid fever is

- a. 1 to 2 days;
- b. 2 to 7 days;
- c. 15 to 20 days;
- d. 8 to 14 days

Answer: (d)

23. Step ladder pattern of fever manifestation is seen in

- a. Pneumonia;
- b. Typhoid;
- c. Malaria;
- d. Tuberculosis

Answer: (b)

24. Salmonella typhi is responsible for the manifestation of

- a. Pneumonia;
- b. Malaria;
- c. Typhoid;
- d. Tuberculosis

Answer: (c)

25. Widal test is confirmatory for the diagnosis of

- a. Pneumonia;
- b. Malaria;
- c. Typhoid;
- d. Tuberculosis

Answer: (c)

26. The available species of plasmodium responsible for malaria manifestation are

- a. 2; b. 3;
- c. 4; d. 6

Answer: (c)

27. Cerebral malaria is caused by

- a. Plasmodium vivax;
- b. Plasmodium malariae;
- c. Plasmodium ovale;
- d. Plasmodium falciparum

Answer: (d)

28. The type of jaundice seen in the patients of malaria is

- a. Prehepatic jaundice;
- b. Hepatic jaundice;
- c. Posthepatic jaundice;
- d. All of the above; e. Any of the above

Answer: (a)

29. Malaria is transmitted by vector

- a. Female anopheles mosquito;
- b. Female culex mosquito;
- c. Aedes aegypti;
- d. Common fly

Answer: (a)

30. The nature of fever observed in malaria is/are

- a. Tertian;
- b. Quartan;
- c. Continuous;
- d. Any of the above

Answer: (d)

Chapter resources

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